

ICS 321: Database Systems (251)

Project # 2

Campus Events Database

(Due Date – December 11, 2025)

A system is to be created to **manage** the calendar of campus events.

1. Each event has a descriptive name and has a single start date and time (when the event begins) and ending date and time.
2. Each event can be classified as sports, social, religious, or academic. Depending on the classification, specific event information is stored.
3. Each event takes place at a specific venue on campus. A venue is either a sports area (such as an athletic field), a lecture hall, a conference hall, or a public space. The venue information differs depending on the type of venue.
4. Each event is sponsored by an academic department. Each academic department has a person responsible for the event.

You have already **modeled** the above requirements in Assignment # 4. Review the following additional requirements:

5. Events can be organized by faculty, students or staff or dependents of faculty or staff.
6. Each event can have multiple sub-events. Each sub-event has a person in-charge reporting to the person responsible for the whole event and maintains the start time of the sub-event and allocated time for the sub-event.
7. An event requires approval from the Events Management Department head. Requests are either approved or rejected with justification for rejection stored in the database. Events are allowed to be deleted before approval/rejection.
8. When an event is approved, a trigger sends an email to maintenance, office services, and security departments.
9. All events can be for a maximum of 3 days duration and must be scheduled between 8 AM and 12 midnight.

Modify the model you have developed to include the additional requirements mentioned above and complete the following steps:

- a. **Design** a complete Conceptual Schema (EER diagram - Enhanced Entity Relationship Diagram) for the above description using Crow's Foot notation. Your diagram should have all the needed details with proper super and sub classes, weak and strong entities defined. You may make any reasonable assumptions, but you must state them clearly.
- b. **Convert** the Conceptual Schema (EER model) to a Logical Schema (set of relations) with all the required attributes. Your relational database must be in BCNF. You must show each step (1NF, 2NF, BCNF).
- c. **Develop** SQL create statements. This should include not only creation of tables, but also (as appropriate), creation of constraints, triggers, and at least one view.
- d. **Populate** your database with sample data to allow testing of the schema and the various transactions using SQL select/insert/update/delete statements as needed. Each table should have a minimum of five rows.
- e. **Use** a stored procedure to set up and populate your database (developed in the previous two steps).
- f. **Test** your work using test queries to ensure the correctness of your design model.

Project 2 **submission** details will be provided later.